

ENTERPRISE INFRASTRUCTURE PLAN

ABC Startup Solutions

**ITEP 414 - System Administration and Maintenance
Week 2 Portfolio Project**

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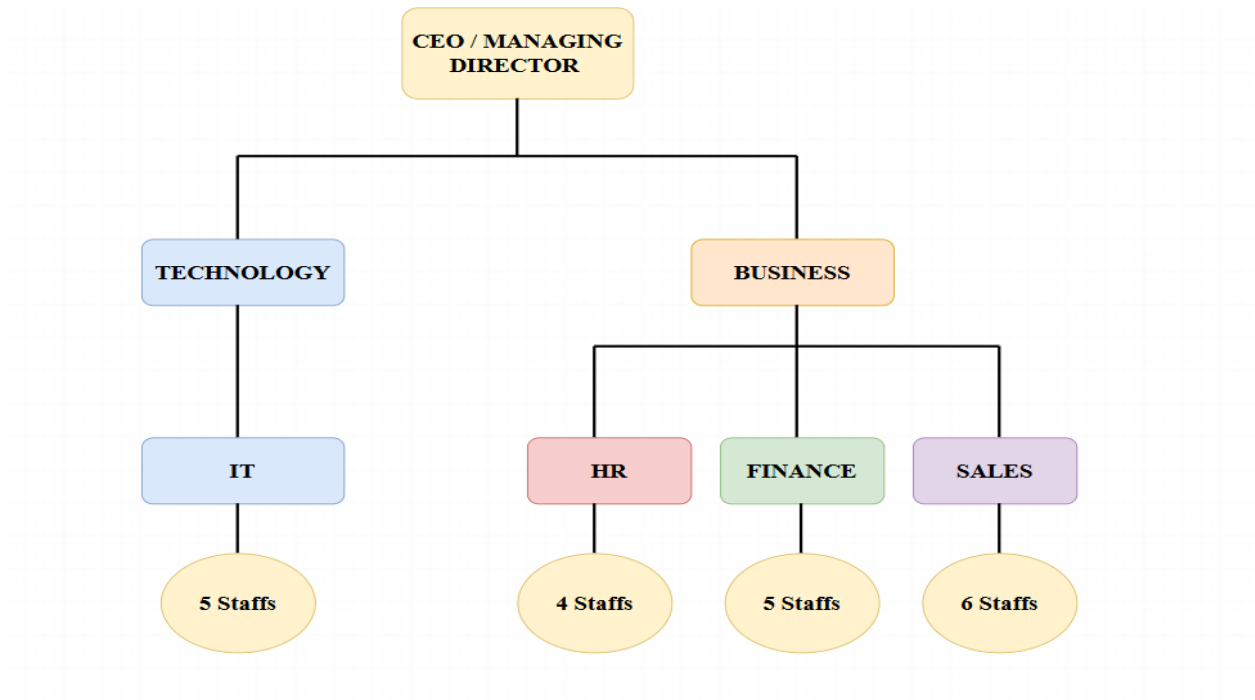
Date: August 14, 2026

Part 1 - Company Profile

1.1 Company Information

Item	Details
Company Name	ABC Startup Solutions
Nature of Business	A software company that develops useful digital solutions for businesses.
Company Vision	To build useful technology that makes work easier.
Company Mission	To create reliable software while keeping company data safe.
Office Location	Magdalena, Laguna, Philippines

1.2 Organizational Structure



1.3 Employee Distribution

Department	Employees
Information Technology	5
Human Resources	4
Finance	5
Sales	6
TOTAL	20

PART 2 - ENTERPRISE HARDWARE INVENTORY

Asset ID	Hardware	Quantity	Department	Purpose
1	Desktop Computer	20	All Department	Main computers for employees
2	Laptop	5	IT / Management	Portable work and system administration
3	Server	1	IT	Stores company services, files, and internal systems
4	Router	1	IT	Manages communication between departments
5	Switch	1	IT	Connects wired computers and network devices
6	Printer	2	HR / Finance	Printing documents and records
7	UPS	6	IT	Protects equipment from power interruptions
8	Wireless Access Point	2	All Departments	Provides wireless network access
9	NAS Storage	1	IT	Centralized file storage and backups
10	External Backup Drive	2	IT	Offline backup of important company data
11	Monitors	20	All Departments	Display for employee workstations

The hardware was selected based on the company's 20 employees and its need for reliable daily operations. Each employee is provided with a desktop computer and monitor, while additional laptops are assigned for mobile work and IT tasks. A server and NAS are included for company services, file storage, and backups. Network devices such as the router, switch, and

wireless access points provide office connectivity. UPS units and external backup drives are also included to help protect important equipment and company data.

PART 3 - ENTERPRISE SOFTWARE INVENTORY

Software	Version	License	Purpose
Windows 11 Pro	Latest Version	Commercial	Main operating system for employee computers
Ubuntu Server	24.04 LTS	Open Source	Operating system for the company server
Microsoft Office	Microsoft 365	Subscription	Used for documents, spreadsheets, and presentations
VS Code	Latest Version	Free	Used by IT staff for coding and development
Git	Latest Version	Open Source	Tracks changes in software projects
Github Desktop	Latest Version	Free	Makes Git and GitHub easier to manage
VirtualBox	Latest Version	Free / Open Source	Used to create and test virtual machines
Google Chrome	Latest Version	Free	Web browsing and accessing online services
Microsoft Defender	Latest Version	Included	Protects computers from malware and other threats
AnyDesk	Latest Version	Commercial / Free Options	Remote support and troubleshooting
7-Zip	Latest Version	Open Source	Compresses and extracts files

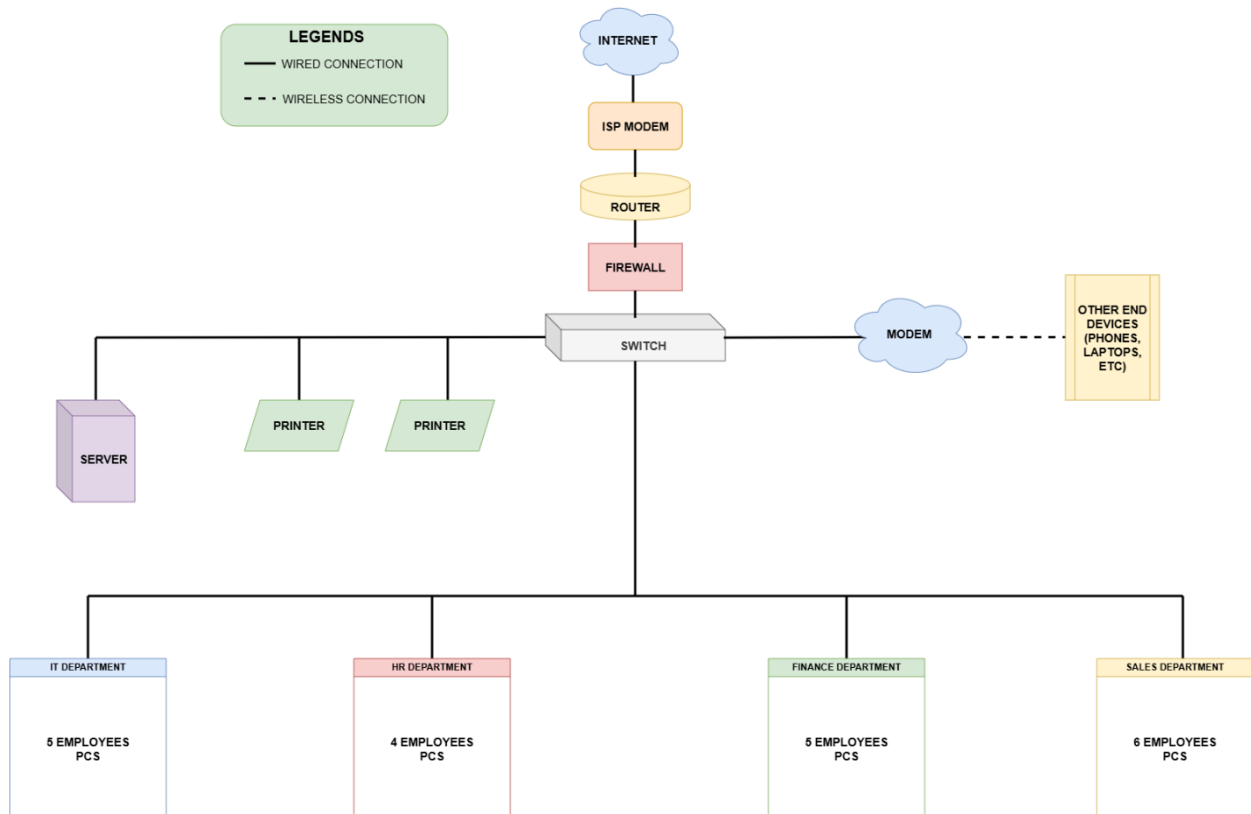
The software was selected to support the daily work of ABC Startup Solutions. Windows 11 Pro and Ubuntu Server provide the main operating systems for the employees and server. Microsoft Office is used for common office tasks, while VS Code, Git, and GitHub Desktop support software development. Security and maintenance tools such as Microsoft Defender, AnyDesk, and 7-Zip are also included to help with protection, support, and file management.

PART 4 - ENTERPRISE NETWORK INVENTORY

Asset ID	Network Equipment	Quantity	Purpose
1	ISP Modem	1	Connects the office network to the internet service
2	Router	1	Manages traffic between the office network and the internet
3	Firewall	1	Helps protect the network from unauthorized access
4	Switch	1	Connects computers, servers, printers, and other wired devices
5	Wireless Access Point	2	Provides Wi-Fi access throughout the office
6	Patch Panel	1	Organizes and manages network cable connections
7	CAT6 Cable	500 meters	Connects wired devices to the network
8	RJ45 Connectors	100 pieces	Used to terminate CAT6 Ethernet cables

The network equipment was selected to provide a stable and secure connection for all departments. The modem provides internet access, while the router manages network traffic and the firewall adds network protection. The switch connects the company's wired devices, while the access points provide wireless connectivity. A patch panel keeps the cables organized, and CAT6 cables and RJ45 connectors are used for the office network connections.

PART 5 - ENTERPRISE NETWORK DIAGRAM



PART 6 - SYSTEM ADMINISTRATION ROLES

6.1 Helpdesk Technician

Responsibilities

Helps employees with common computer and software problems. They troubleshoot issues, set up computers, reset accounts, install software, and assist users with basic technical concerns.

Skills

- Basic computer troubleshooting
- Windows and software knowledge
- Good communication
- Problem-solving
- Customer service

Common Tools

- Windows
- AnyDesk
- Microsoft Defender
- Remote support tools
- Ticketing systems

Certifications

- CompTIA A+
- Cisco Certified Support Technician (CCST)

6.2 Network Administrator

Responsibilities

Manages the company's network and makes sure employees can connect to company resources and the internet. They configure network devices, monitor connections, troubleshoot network problems, and help keep the network secure.

Skills

- Networking fundamentals
- IP addressing
- Router and switch configuration
- Troubleshooting
- Network security

Common Tools

- Cisco IOS
- Wireshark
- Ping and other network commands
- Network monitoring tools
- Router and switch management tools

Certifications

- Cisco Certified Network Associate (CCNA)
- CompTIA Network+

6.3 Linux System Administrator

Responsibilities

Manages Linux servers and makes sure they are running properly. They manage users, install software, configure services, monitor systems, and handle security and backups.

Skills

- Linux command line
- User and permission management
- Server maintenance
- Shell scripting
- Troubleshooting
- Basic security

Common Tools

- Ubuntu Server
- SSH
- Bash
- Systemctl
- Linux terminal

Certifications

- Red Hat Certified System Administrator (RHCSA)
- Red Hat Certified Engineer (RHCE)

6.4 Cloud Administrator

Responsibilities

Manages cloud-based systems and services. They help set up cloud resources, manage access, monitor systems, handle backups, and make sure cloud services are secure and available.

Skills

- Cloud computing
- Networking
- Security
- Storage management
- Virtual machines
- Monitoring and troubleshooting

Common Tools

- AWS
- Microsoft Azure
- Azure Portal

- AWS Management Console
- Cloud monitoring tools

Certifications

- AWS Certified Cloud Practitioner
- Microsoft Certified: Azure Administrator Associate

6.5 How the Four Professionals Work Together

The four professionals work together to keep the company's IT systems running properly. The Helpdesk Technician assists employees with daily technical problems and handles basic troubleshooting. The Network Administrator manages the company's network and makes sure devices can communicate properly. The Linux System Administrator takes care of Linux servers and their services, while the Cloud Administrator manages cloud-based resources and services. When a problem involves more than one area, they can work together to find the cause and fix it without affecting the company's operations.

PART 7 — INFRASTRUCTURE RECOMMENDATIONS

7.1 Internet Provider

Recommended: A reliable business internet plan with at least 200 Mbps download speed.

ABC Startup Solutions should use a reliable internet provider with a business plan. A 200 Mbps connection should be enough for 20 employees who need internet access for communication, software development, cloud services, and online meetings. A business plan is also better because it can provide better support when connection problems happen.

7.2 Server Specifications

Recommended Server:

Component	Recommendation
Processor	Intel Xeon or similar
RAM	32 GB
Storage	1 TB SSD
Operating System	Ubuntu Server LTS
Network	Gigabit Ethernet

7.3 Backup Strategy

ABC Startup Solutions should use a 3-2-1 backup strategy. Important company files should have three copies, stored on two different types of storage, with one copy kept offline or away from the main office. The NAS can be used for regular backups, while the external backup drives can be used for offline copies. Backups should be done daily, with regular checks to make sure the files can actually be restored.

7.4 Security Recommendations

The company should use a firewall, strong passwords, regular software updates, user access controls, and backups. Employees should only have access to the files and systems they need for their jobs. The network should also be monitored by the IT department so unusual activity can be noticed early.

7.5 Antivirus

Microsoft Defender can be used on the company's Windows computers because it is already included with Windows 11. It provides basic protection against malware and other common threats. The IT department should make sure that Defender is enabled and regularly updated on all company computers.

7.6 Password Policy

- Use strong and unique passwords.
- Do not share passwords with other employees.
- Use multi-factor authentication when available.
- Change passwords when an account may have been compromised.
- Lock accounts after repeated failed login attempts.

7.7 Expansion Plan

The network should be designed so the company can add more employees and devices later without replacing the whole setup. The managed switch should have extra ports, and the server and storage should have room for future upgrades. The company can also add more wireless access points, computers, storage, or cloud services as the business grows.

7.8 Overall Recommendation

The recommended infrastructure is designed to meet the current needs of ABC Startup Solutions while leaving room for future growth. Reliable internet, a properly sized server, regular backups, network security, and a clear password policy can help the company keep its systems stable and protect its data. The infrastructure can also be expanded as more employees and devices are added.

PART 8 – PERSONAL REFLECTION

I actually learned a lot from this project, especially about how hard and a bit complicated planning an enterprise infrastructure can be. Before doing this activity, I thought it was mostly about computers and networks, but I realized that there are many things that need to be considered before everything can work properly.

One thing I found interesting is how an enterprise infrastructure lets you see the whole setup of a company. For me, it is like a blueprint because you can see the different parts of the system and how they connect with each other. It gives you a better idea of how everything is supposed to work before actually setting it up.

The task that was the most challenging for me was Part 5, which was the network diagram. I had to think more carefully about where each device should go and how everything should connect. I couldn't just put the devices anywhere because the connections needed to make sense. I had to think about the internet, modem, router, firewall, switch, server, printer, wireless access point, and the different departments. Compared to the other parts, this was the one that took the most thinking for me.

I also realized why planning is important before deploying an IT infrastructure. For me, it is like building a house. The infrastructure plan is like the foundation of the house. If the foundation is weak or poorly planned, the whole thing might not be stable and could cause problems later. In the same way, a company needs proper planning so its IT systems can be reliable, secure, and easier to manage.

This project can help me become a better System Administrator because it taught me to think ahead instead of just solving problems after they happen. It also helped me understand that a System Administrator needs to know how different parts of an IT infrastructure work together. Overall, this activity gave me a better idea of what system administration is really like and why proper planning is important before setting up any IT system.

